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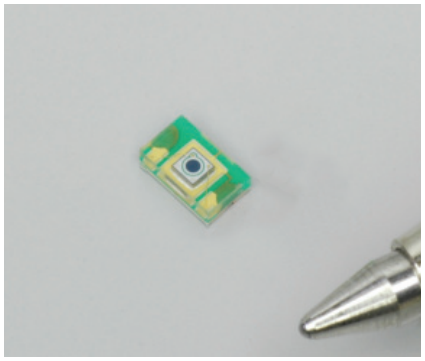
PHOTON IS OUR BUSINESS

Si APD



Low bias operation, for 800 nm band, small package

Features

- Miniature and thin package: 1.8 × 3.1 × 1.0^t mm
- Stable operation at low bias
- High-speed response
- High sensitivity
- Low noise

Applications

- Optical rangefinder
- Laser radar
- FSO (free space optics)

General ratings

Parameter	Symbol			Unit
Active area size*1	A		ϕ0.5	mm
Effective active area	-		0.19	mm^2
Package	-	Plastic		-

*1: Active area in which a typical gain can be obtained

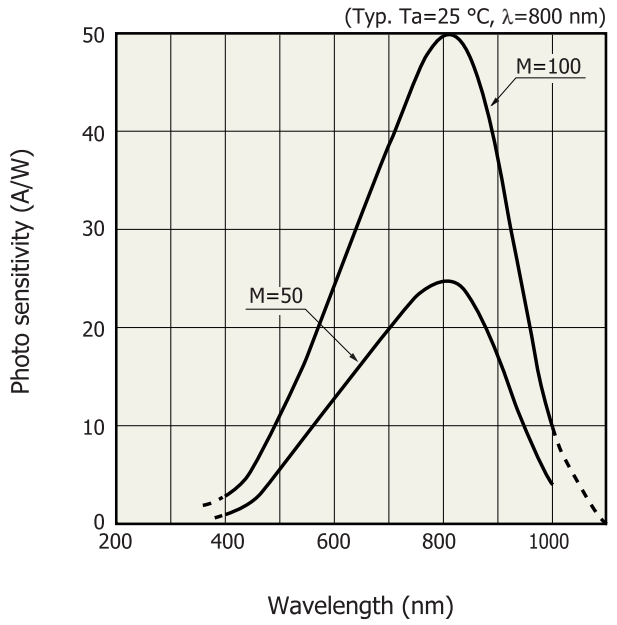
Absolute maximum ratings

Parameter	Symbol	Value	Unit
Operating temperature	Topr	-20 to +60	°C
Storage temperature	Tstg	-40 to +80	°C
Reverse current	Irmax	200	μA
Forward current	Ifmax	10	mA

Electrical and optical characteristics (Ta=25 °C)

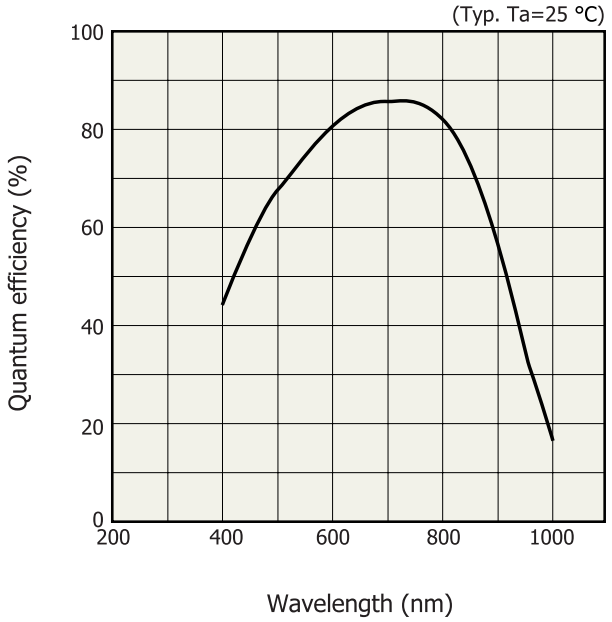
Parameter	Symbol	Condition				Min.	Typ.	Max.	Unit
Spectral response range	λ					400 to 1000			nm
Peak sensitivity wavelength	λp	M=100				-	800	-	nm
Photo sensitivity	S	λ=800 nm, M=1				-	0.5	-	A/W
Quantum efficiency	QE	λ=800 nm, M=1				-	75	-	%
Breakdown voltage	V_{BR}	I_D=100 μA				-	150	200	V
Temperature coefficient of V_{BR}	-					-	0.65	-	V/°C
Dark current	I_D	M=100				-	100	1000	pA
Cut-off frequency	fc	M=100, R_L=50 Ω λ=800 nm, -3 dB				-	900	-	MHz
Terminal capacitance	Ct	M=100, f=1 MHz				-	2	-	pF
Excess noise figure	x	M=100				-	0.3	-	-
Gain	M	λ=800 nm				-	100	-	-

Spectral response



KAPDB0020EC

Quantum efficiency vs. wavelength



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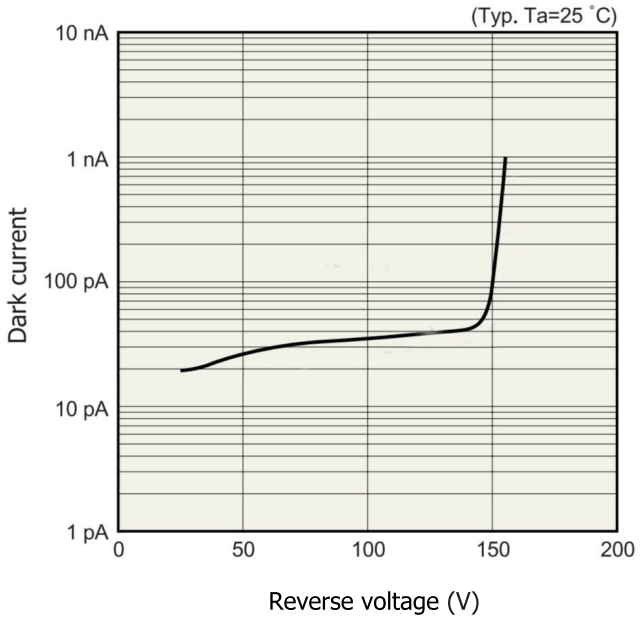
Dark current vs. reverse voltage



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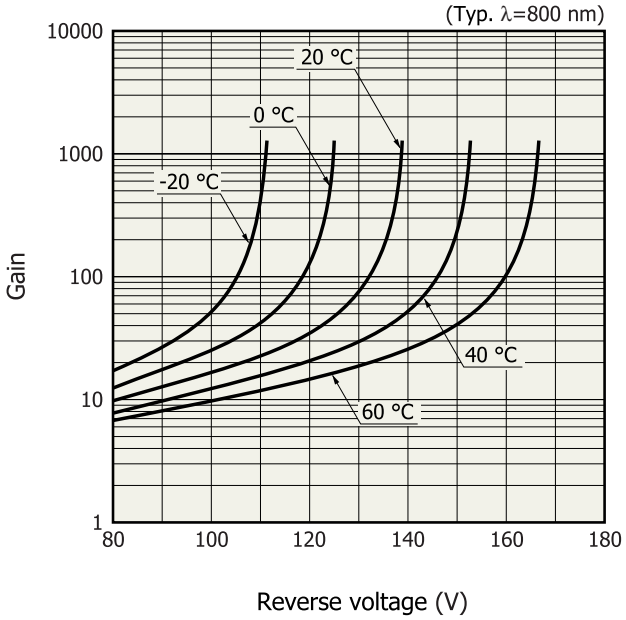
Gain vs. reverse voltage



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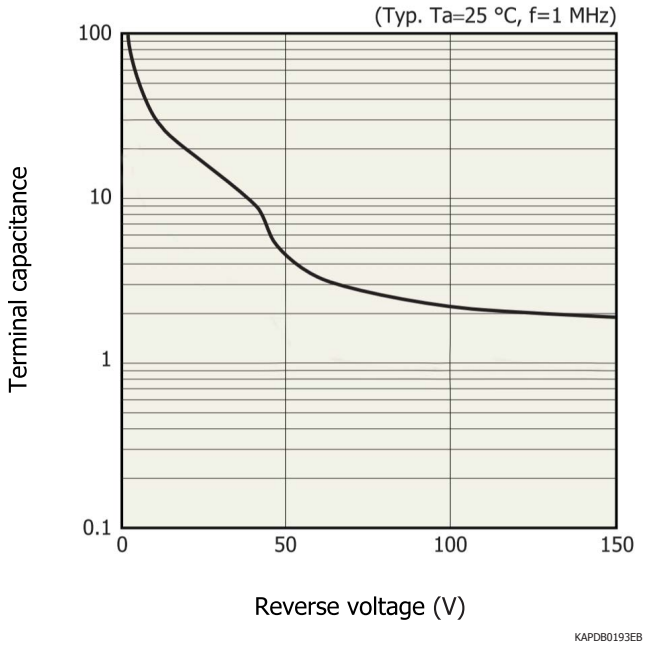
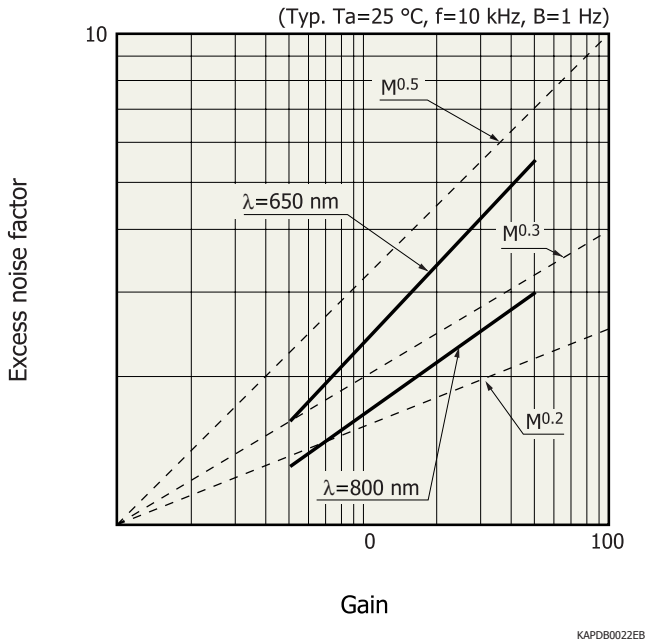
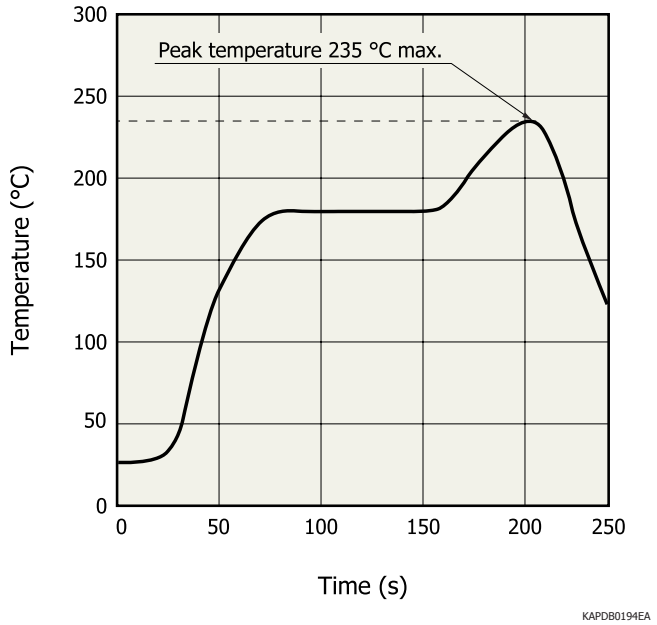
Terminal capacitance vs. reverse voltage



Excess noise factor vs. gain

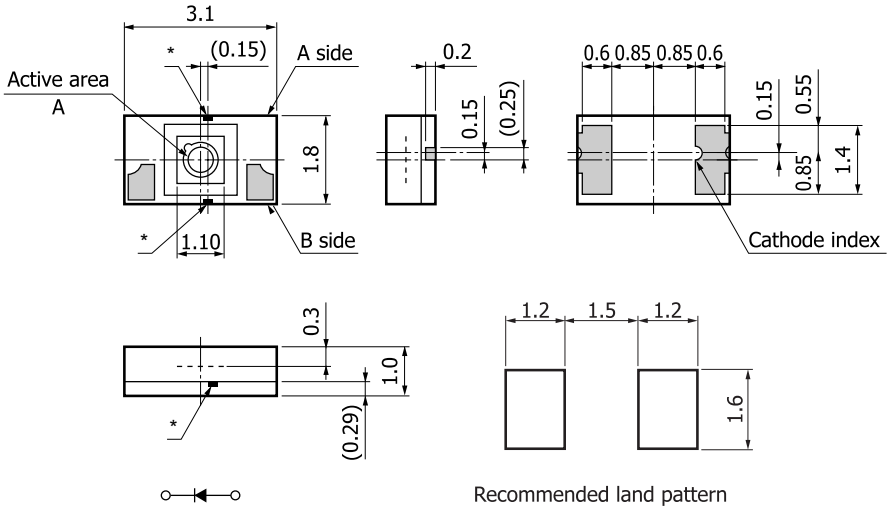


Recommended solder reflow condition



- After unpacking, store this device in an environment at a temperature of 30 °C and a humidity below 60%, and perform reflow soldering on this device within 24 hours.
- Thermal stress applied to the device during reflow soldering differs depending on the PC boards and reflow oven being used.
- When setting the reflow conditions, make sure that the reflow soldering process does not degrade device reliability.

Dimensional outline (unit: mm, tolerance unless otherwise noted: ±0.2 mm)



Active area position accuracy: X, Y≤±0.3
Values in parentheses indicate reference value.
* Package side
Wiring is exposed on A and B sides.
Do not allow any conductor to make contact with the package sides to avoid shorting.
Electrodes

Type no.	A
Si APD	φ0.5

KAPDA0130EA

Information described in this material is current as of May, 2011. Product specifications are subject to change without prior notice due to improvements or other reasons. Before assembly into final products, please contact us for the delivery specification sheet to check the latest information.

The product warranty is valid for one year after delivery and is limited to product repair or replacement for defects discovered and reported to us within that one year period. However, even if within the warranty period we accept absolutely no liability for any loss caused by natural disasters or improper product use.
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